David Bedolla, PhD

Robotics Controls | Trajectory Planning | Whole-Body Control & Mobile Manipulation

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EXPERIENCE

Associate Researcher
Mar 2026 - Present | Lab INIT Robots, Montréal

- Adapted my swivel-angle method to the Kinova Gen3 joint-velocity interface for Cartesian tracking and elbow posture control on an AgileX mobile base.
- Designed the single-joystick workflow for local and world-frame commands. Reviewed AI-generated C++ and integrated and tested the ROS 2 system on hardware.
- Compared PID and super-twisting feedback during base rotation and observed improved target tracking with super-twisting.

Postdoctoral Researcher
Mar 2025 - Mar 2026 | École de technologie supérieure (ÉTS), Montréal

- Cut the cable-driven exoskeleton's complete LabVIEW read-compute-write cycle from about **60 ms to 2 ms** by moving motor setup out of the loop and removing repeated calculations.
- Implemented fixed-time sliding-mode torque control and tested hip and knee motion against a walking trajectory on the physical exoskeleton.
- Reduced steering oscillations in autonomous-vehicle tests by improving ROS/Python control scheduling.
- Built symbolic dynamics tools and supported international assistive robotics teams with 6-DoF inverse kinematics and real-time motion control.

Research Professor
Mar 2024 - Mar 2025 | Universidad Tecnológica de la Mixteca, Mexico

- Implemented and tested online particle swarm gain tuning on a physical 7-DoF exoskeleton in Montréal. The co-authored study reported **over 20% lower tracking error** than heuristic tuning.
- Reviewed the manuscript for a co-authored study on a three-finger adaptive gripper.

Research Assistant
Jan 2020 - Dec 2023 | École de technologie supérieure (ÉTS), Montréal

- Implemented and tested robust and learning-based model predictive control (MPC) in LabVIEW Real-Time on the physical 7-DoF ETS-MARSE exoskeleton.
- Reported **387 µs** for 10-step robust MPC and **687 µs** for a separate learning-based MPC experiment using a Gaussian process residual model.
- Developed analytical inverse kinematics with a learned elbow swivel reference and combined EMG and IMU acquisition in C++ for mirror-rehabilitation experiments.

Research Professor
Oct 2018 - Oct 2019 | UNSIJ, Oaxaca, Mexico

- Implemented real-time neural-network inverse kinematics for a 6-DoF PUMA manipulator.

Professor
Feb 2017 - Oct 2018 | TecNM, Puebla, Mexico

- Co-authored the 2018 paper on the embedded manipulator emulator developed during my master's research.

TECHNICAL SKILLS

Controls & Planning Trajectory planning, whole-body control, swivel-angle redundancy resolution, MPC, sliding-mode control, torque control

Robotics Software C++, ROS 2, Python, MATLAB/Simulink, CMake, Qt, Pinocchio, URDF, Ruckig

Real-Time Systems LabVIEW RT/FPGA, NI real-time targets, TI C2000, embedded C, analog I/O, DAQ, EMG/IMU acquisition

Modeling & Simulation Kinematics, rigid-body dynamics, Gazebo, MuJoCo, Isaac Sim

Robot Learning Gaussian processes, neural-network inverse kinematics, PyTorch reinforcement learning in simulation

SELECTED ROBOTICS PROJECTS

Whole-Body Mobile Manipulation
Used base odometry to transform a world-frame target into the arm frame. Tested arm compensation during base rotation and adjusted damping and speed limits to reduce visible overshoot.
Mobile manipulator video | Kinova Gen3 video

PlannerGUI for URDF-Based Trajectory Planning
Designed the C++/Qt planning workflow, reviewed AI-generated code, and specified diagnostics to make failed inverse-kinematics samples visible. A logged Kinova Gen3 desktop run converged on all 5,001 offline samples in 84.93 ms for a five-second path.

ETS-MARSE Human-Like Motion and Mirror Rehabilitation
Used wearable motion measurements and learned elbow posture to generate joint references for mirror exercises. The physical setup used a 1 ms LabVIEW Real-Time period and a 50 µs FPGA current-loop period.
Exoskeleton video

Embedded Hardware-in-the-Loop Manipulator Emulator
Developed a two-joint simulated plant on a TI F28377S with physical analog I/O to an external controller. Published plant update periods were 10 µs with Euler and 30 µs with fourth-order Runge-Kutta. HIL paper

Tactile Contact Maps in Gazebo and MuJoCo
Specified a shared ROS 2 interface for simulated TSF-85 contact maps, reviewed AI-generated adapters, and tested contact locations on the gripper in both simulators.

EDUCATION

PhD in Robotics | 2020 - 2023
École de technologie supérieure (ÉTS), Montréal
Dissertation *Learning-based upper limb robotic rehabilitation*

M.Sc. in Robotics | 2014 - 2016
Universidad Tecnológica de la Mixteca
Thesis on hardware-in-the-loop simulation of a robotic manipulator

B.Eng. in Mechatronics | 2009 - 2014
Universidad Tecnológica de la Mixteca | CENEVAL EGEL Outstanding distinction

Recognition 2024 ÉTS Excellence Award nomination, 2020 doctoral scholarship, 2014 master's scholarship

SELECTED PUBLICATIONS

Sosa Méndez, D., **Bedolla-Martínez, D.**, Saad, M., Kali, Y., García Cena, C. E., and Álvarez, Á. L. (2025).
Upper-Limb Robotic Rehabilitation: Online Sliding Mode Controller Gain Tuning Using Particle Swarm Optimization.
Robotics, 14(4), 51. DOI 10.3390/robotics14040051

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023).
Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation.
Engineering Applications of Artificial Intelligence, 126, 106966. DOI 10.1016/j.engappai.2023.106966

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023).
Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm.
IEEE PEDS. DOI 10.1109/PEDS57185.2023.10246735